

Bitwise differential trail through **QALQAN** found by SAT

CiVerLy

August 21, 2026

1 QALQAN

Maximal differential probability: 2^{-38}

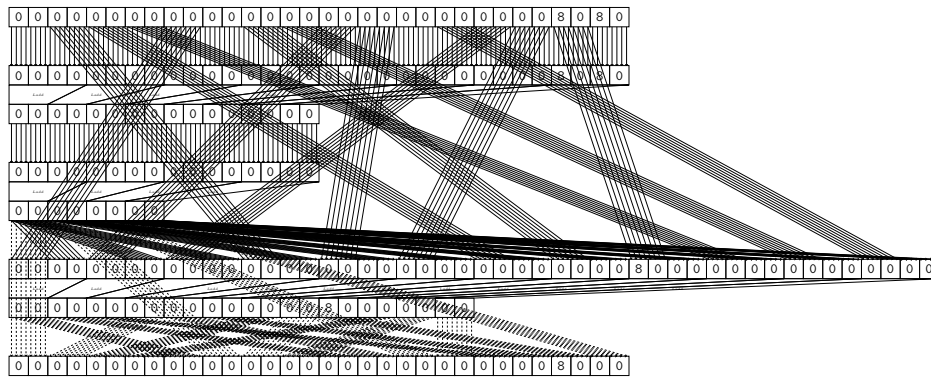
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
KrychAbstart																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
B-Bin-Layer																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	8	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	8	0	
L																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	
A4E1D9																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	1		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	1		
B-Bin-Layer																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0		
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L																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0		
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A4E1D9																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0		
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B-Bin-Layer																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0		
L																																	
0	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	0	8	0	0	0	0	8	0	8	0					
B-Bin-Layer																																	
0	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	8	0	0	0	0	8	0	8	0						

2 SBoxLayer

Maximal differential probability: $2^{-12.0}$ [illegible]

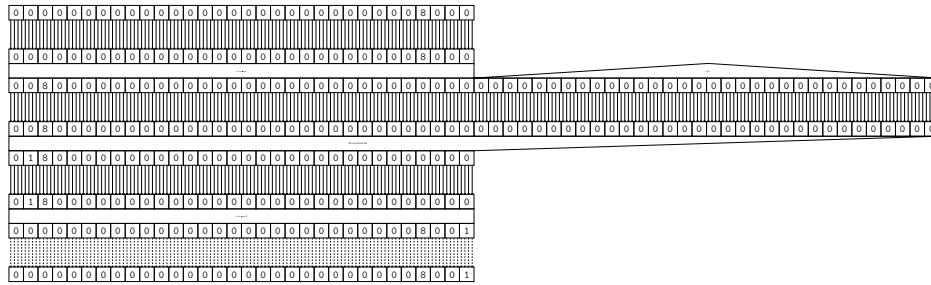
3 L

Maximal differential probability: $2^{-0.0}$



4 Add128

Maximal differential probability: $2^{-2.0}$

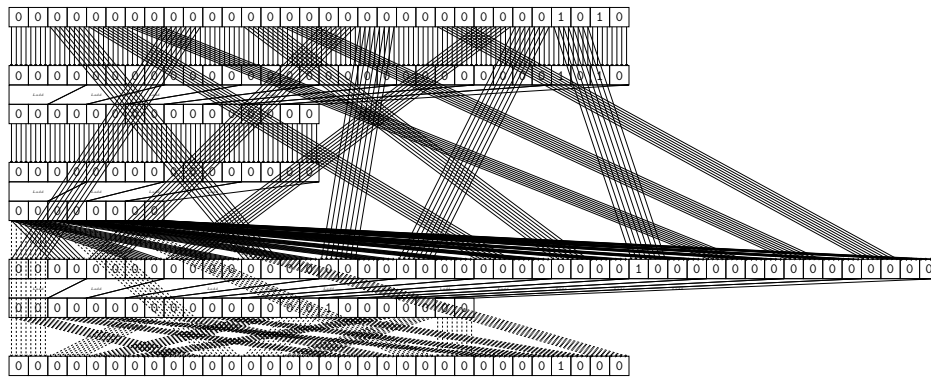


5 SBoxLayer

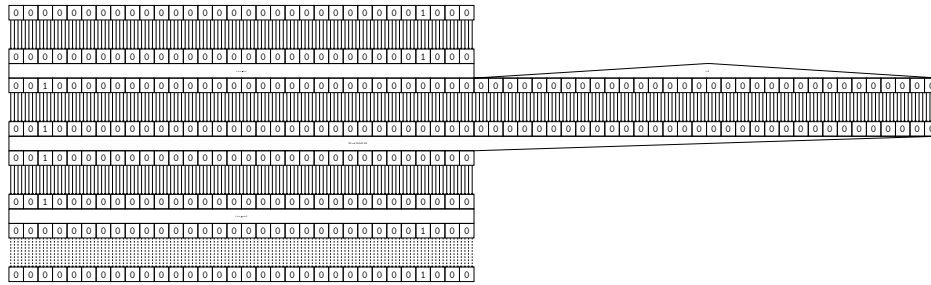
Maximal differential probability: $2^{-14.0}$ [illegible]

6 L

Maximal differential probability: $2^{-2.0}$



7 Add128

Maximal differential probability: $2^{-1.0}$ 

8 SBoxLayer

Maximal differential probability: $2^{-7.0}$ [illegible]

9 L

Maximal differential probability: $2^{-0.0}$

